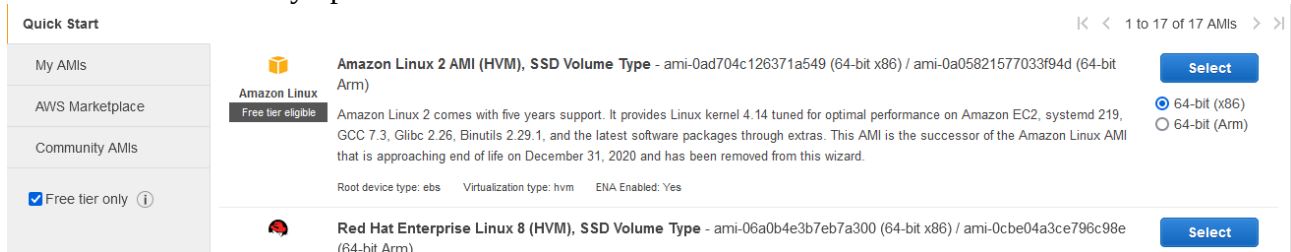


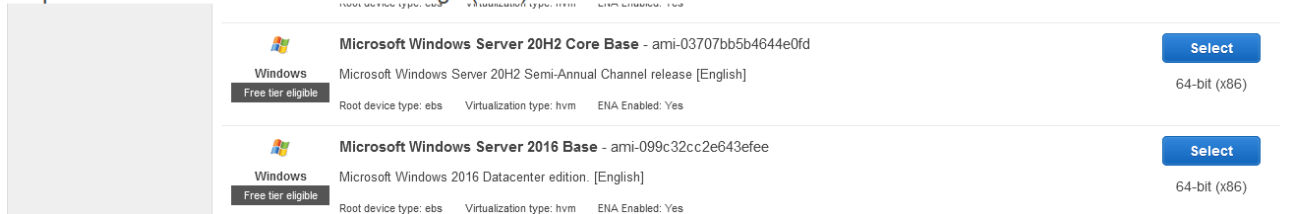
Creating a Windows EC2 Instance

1. Go to console.aws.amazon.com and login with your credentials.
2. Select EC2 service or click Launch a Virtual Machine with EC2 option.
3. Click Free Tier only option on the left hand side.



4. Select Windows Server 2016 Base AMI.

Step 1: Choose an Amazon Machine Image (AMI)



5. Keep the default **t2.micro** instance type. Click Next: Configure Instance Details
6. Keep all the default options on the Instance details page. Click Next: Add Storage button.
7. Keep the default options on the Add storage page (i.e. 30 GB storage). Click Next: Add Tags
8. Do not add any tag. Click Next: Configure Security Group button.
9. To access Windows instance, remote desktop service is used. It uses TCP port 3389. Thus the security group shows entry for the port 3389.

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☒ Create a new security group
☐ Select an existing security group

Security group name:

Description:

Type	Protocol	Port Range	Source	Description
RDP	TCP	3389	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop

Add Rule

10. Click Review and Launch.
11. On the Review Launch Instance page, check the configured options. Finally click Launch button.

12. A window to create a new key will open. Select the create a new key option and specify a name for this new key. Then Click Download Key Pair. Save it to a separate folder. Now click the Launch Instance Button.

Select an existing key pair or create a new key pair

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair

Key pair name

new-win-instl

Download Key Pair

You have to download the **private key file** (*.pem file) before you can continue. Store it in a **secure and accessible location**. You will not be able to download the file again after it's created.

Cancel

Launch Instances

13. Now click on the instance ID shown.

Your instances are now launching

The following instance launches have been initiated: i-0ff6f561437480fe3 [View launch log](#)

14. Once the instance is running, select the check box in front of the Windows EC2 instance. In the details page below, note down the public IPv4 address of the instance.

15. On the same page click the connect button displayed above the instance.

Instances (1/1) Info			Connect	Instance state ▼	Actions ▼	Launch instances	▼
Filter instances		search: i-0ff6f561437480fe3 X Clear filters					
☒	Name ▼	Instance ID ▲	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone
☒	-	i-0ff6f561437480fe3	Running	t2.micro	Initializing	No alarms +	ap-south-1a

16. In the Window that opens click RDP Client option.

Connect to instance Info

Connect to your instance i-0ff6f561437480fe3 using any of these options

Session Manager

RDP client

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following details:

Public DNS

ec2-65-0-100-19.ap-south-1.compute.amazonaws.com

User name

Administrator

Password

Get password

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

17. On that page, Click **Download remote desktop file** button. Save the file to the directory where you saved the key.

18. In the same window click **Get password** option in front of password field below.

19. A new window opens as shown below.

To decrypt the password, you will need your key pair for this instance.

 **Key pair associated with this instance**
new-win-inst

Browse to your key pair:

 **Browse**

Or copy and paste the contents of the key pair below:

Cancel **Decrypt Password**

20. In the above window click **Browse** button. Now select the key that was created for this instance. After selecting the key it will look like as shown below.

 **Key pair associated with this instance**
new-win-inst

Browse to your key pair:

 **Browse**

 **new-win-inst.pem**
1.7KB

Or copy and paste the contents of the key pair below:

```
-----BEGIN RSA PRIVATE KEY-----
MIIeowIBAAKCAQEAwk/YErE/pKEXeovAK5Aw5Uw32gTelbzA14mWMKRLRg100Gqb
DRkXyFRvji1KpIYQqzYUj+a/WWwcL/M3T4Y1haNhms5rm5Gfu70bEzPyp17eHwV
4QUvnVPIY4RcmiJ1vuVaulKFDP7VvMsTxITP6ZLkWDkaZjietyp1cOqSNuQK1/Rt
bl7WzFZ1ccFduzCSxwCIHNvKNEluEtzfbO7TkHwTVM6ULw9d+AOszYwIxm3u6TZN
TvtPPFdH3B8/S4/Dq7YQaVdah9WQFsDUo5iuUAlxwMu69I6j+n7Azpx0pv7ocX/v
yj3A9dIG9s/FMVGU5l438NqjRcGqYCL4GbP8gwIDAQABAolBABmievkMVT8c2YIb
ScD1ht3ZTKz+82c0BaTpy9yi8CNPxydvvyQDTXKXbLdRt+Tj3TPkK4IcJSSNA9SQK
ybYuaQxlcUpYfwWY0X2RfMGE6KzsVcuxYxmKm5t/4/VbEvRqs2n5zBxkopTyUbrU
-----
```

Cancel **Decrypt Password**

21. Now click on **Decrypt Password** button below. A password will be displayed. Copy that password and store securely in a file. Click cancel to close this window.

Session Manager | **RDP client**

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

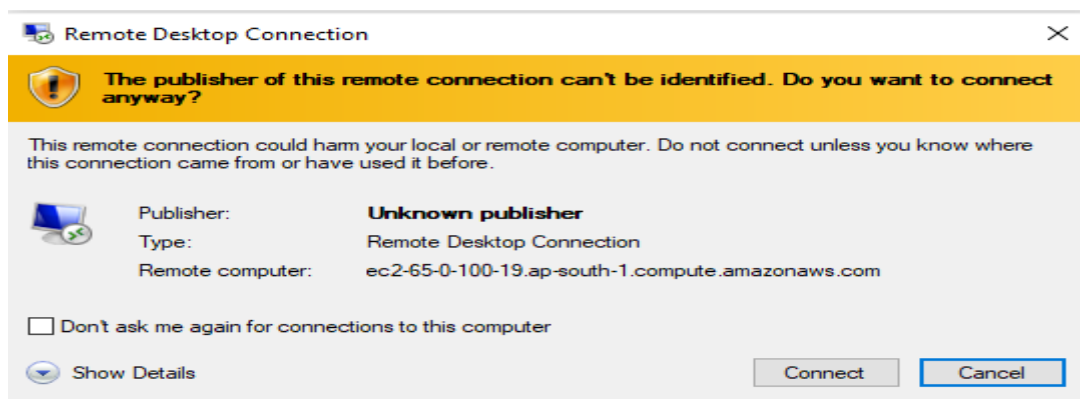
Download remote desktop file

When prompted, connect to your instance using the following details:

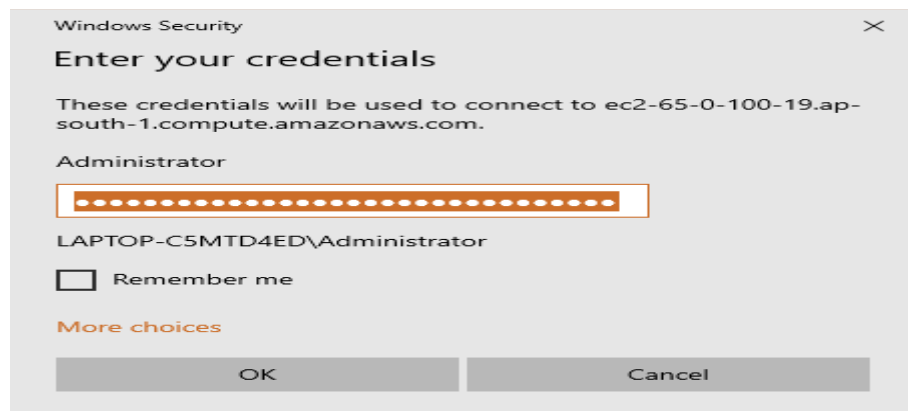
Public DNS	User name
 ec2-65-0-100-19.ap-south-1.compute.amazonaws.com	 Administrator
Password	
 <u>DqBocpx4j3NpT(;Eg*\$8jJO)O;HVmfVf</u>	

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

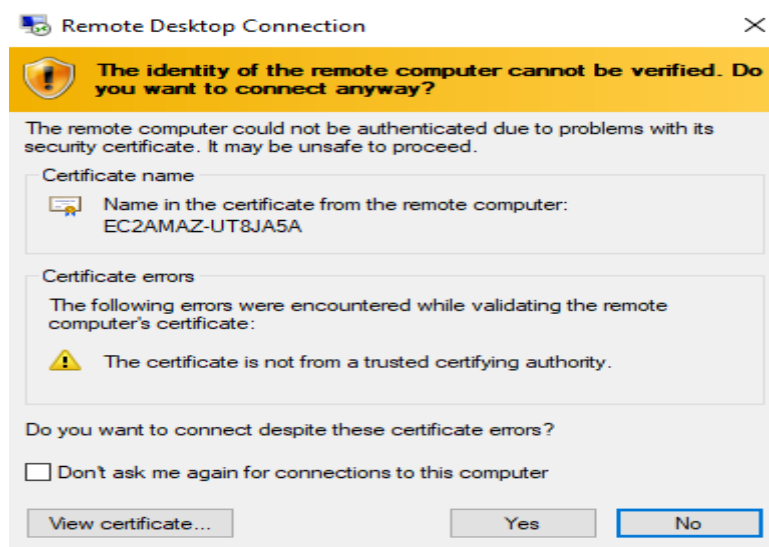
22. Now on your laptop. Go to the directory where the connection file is downloaded in step 17.
23. Double click on this file. It will open a remote desktop client window. Click connect.



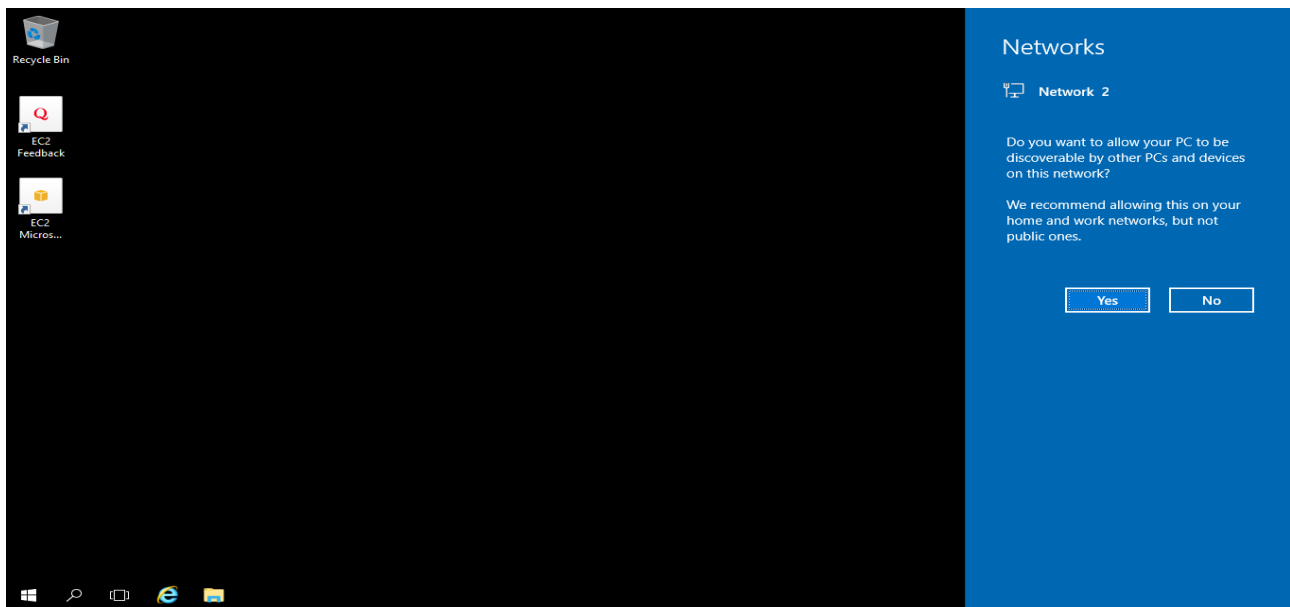
24. It will prompt for the password. Copy the password generated and copied into a file in step 21 and paste in this window. Click OK.



25. The certificate message is displayed as shown below. Click Yes.



26. Now a screen will open that displays the desktop of the Windows server 2016 EC2 instance. Click Yes in the Networks pane.



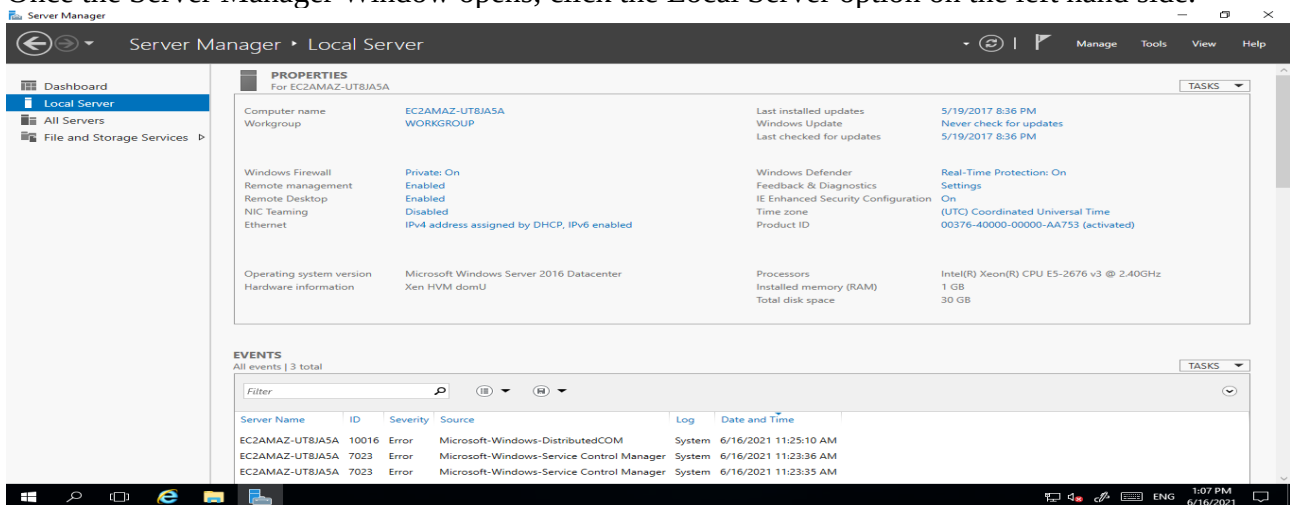
27. Now you have successfully created a Windows EC2 instance and connected to it. Now you can install all required software on this EC2 instance.

Creating a Web site in Windows EC2 Instance

The steps below install open-jdk and tomcat server on the above EC2 instance.

Perform all the below steps in the remote desktop screen of the EC2 instance.

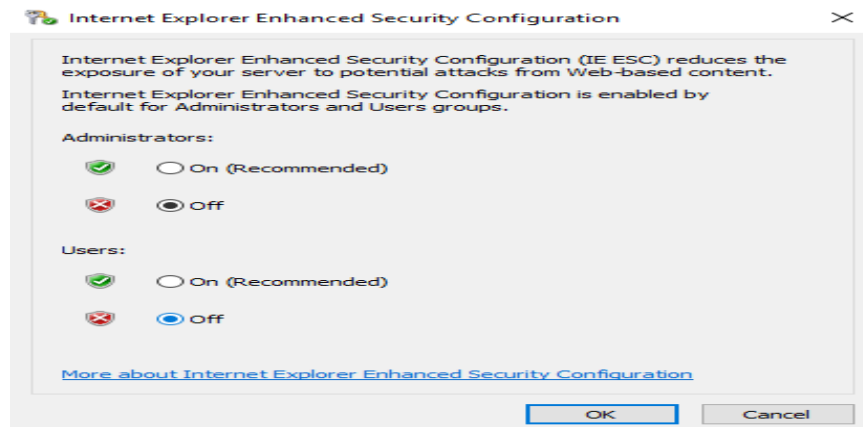
1. first we need to reduce the Internet Explorer security. On Server the IE security is enhanced, which creates a problem while downloading packages. For this click the search button of the screen on the left side. Then in the search box type Server Manager. Click on the Server Manager above. Once the Server Manager Window opens, click the Local Server option on the left hand side.



2. In the above window, on the right hand side check for **IE Enhanced security configuration**.



3. Click on the On. Click off for both administrator and user. Then click ok.

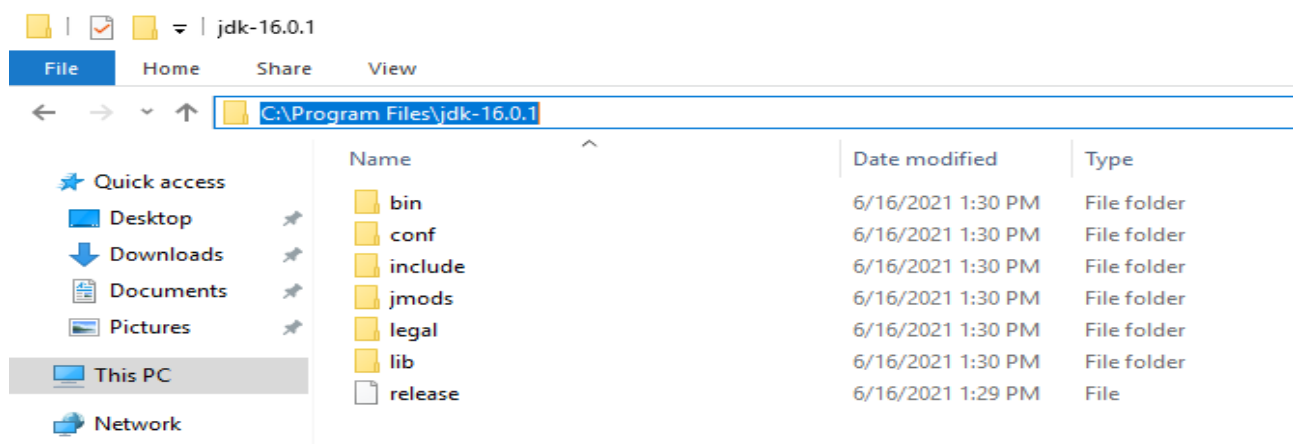


4. Now open Internet Explorer. Go to the website <https://jdk.java.net/>. Click on JDK16 to download. On the page that opens, click zip in front of Windows/x64.

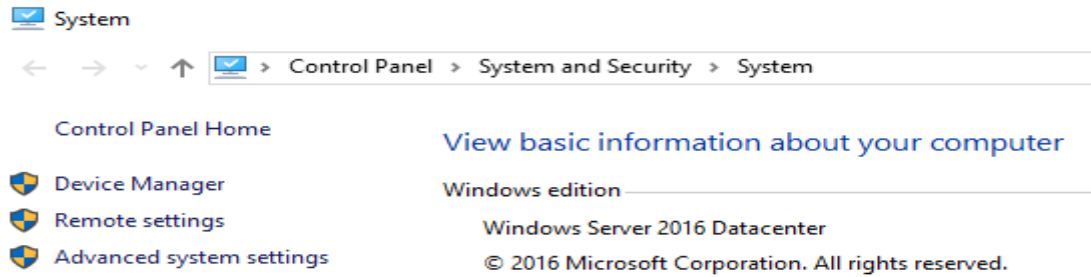
Builds

Linux/AArch64	tar.gz (sha256)	174498972 bytes
Linux/x64	tar.gz (sha256)	184332972
macOS/x64	tar.gz (sha256)	181628041
Windows/x64	zip (sha256)	183657432

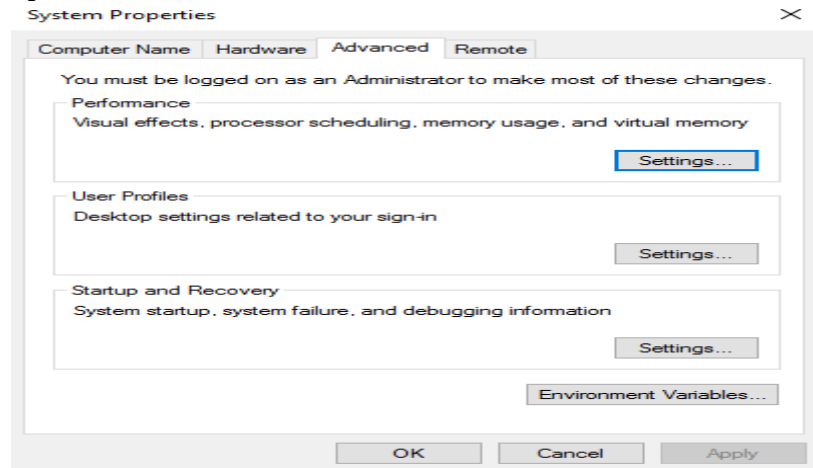
5. Click Save. Then click open folder option in the internet explorer option or open windows explorer and go to Downloads folder. Double click on the zip file downloaded and copy the **jdk-16.0.1** folder to the c:\program files directory. Double click on the folder and click on the path tab to display the path as shown below. Copy the path.



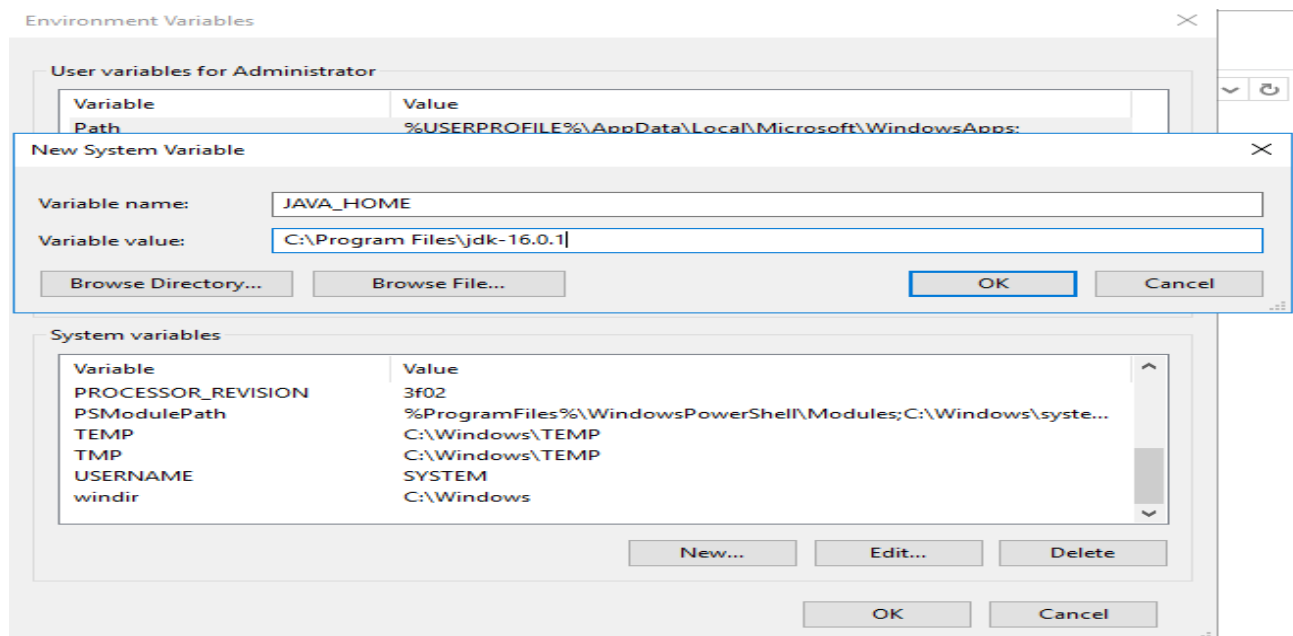
6. In the above window, right click **This PC** option and click properties. In the new window that opens click Advanced system settings on the left side.



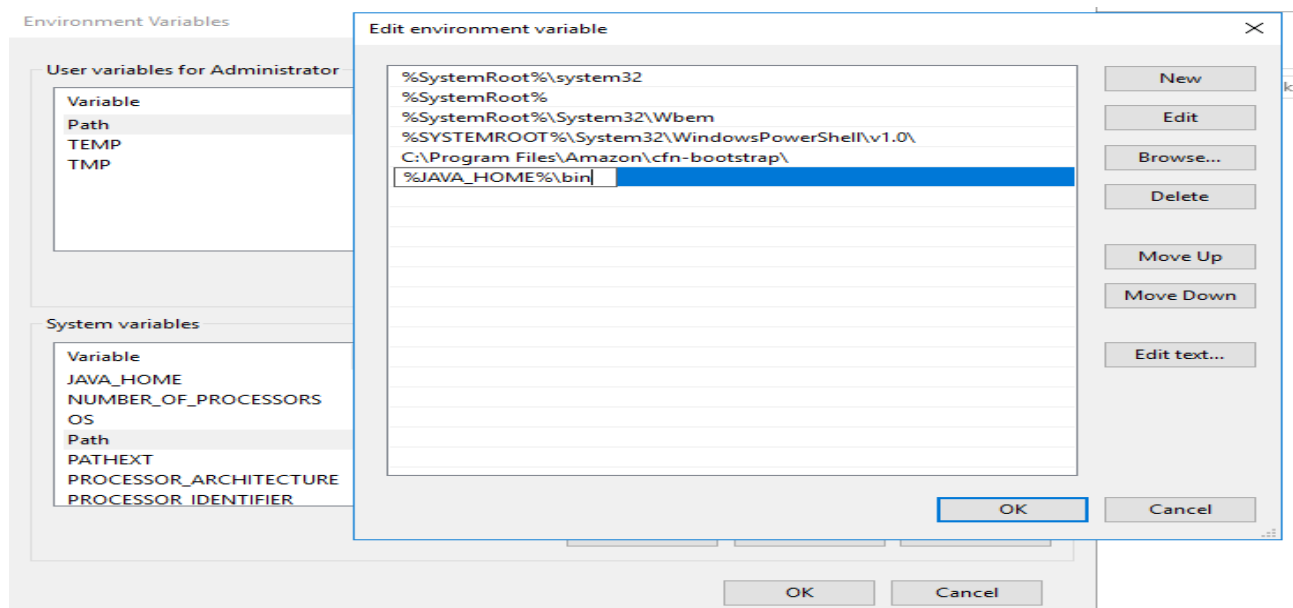
7. In the system properties window click Environment Variables button below.



8. In the window that opens, in the system variable section click New and add JAVA_HOME as the variable name and in the variable value paste the path of the jdk directory. Click OK.

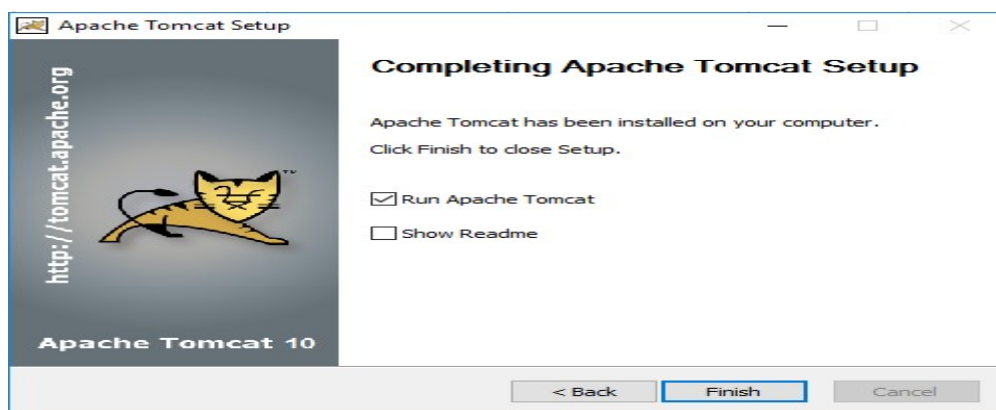


9. Now in the system variable select PATH variable and click Edit. Then click Edit. In the new window that opens, click New and add %JAVA_HOME%\bin. Then Click OK on all windows to close.

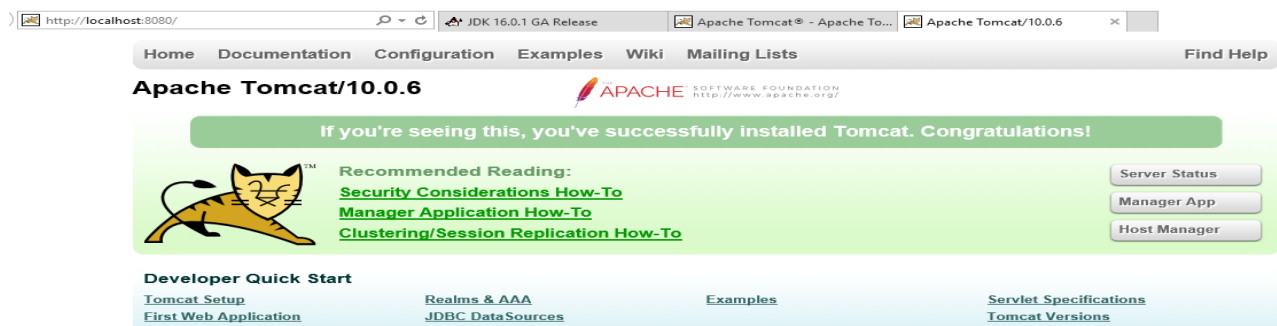


10. Now go to Internet explorer and go to <https://tomcat.apache.org/download-10.cgi> . Then scroll down and download **32 bit/64 bit Windows Service Installer**. Save the file.

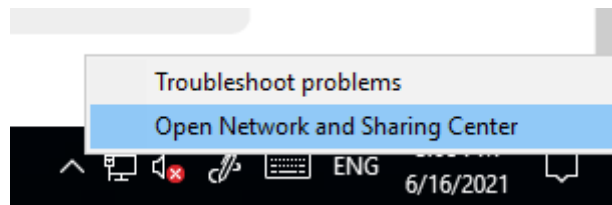
11. Once the download is complete, go the Downloads folder and double click on the installer. Click the default option or next to install the tomcat server. On the last screen uncheck show readme and click finish.



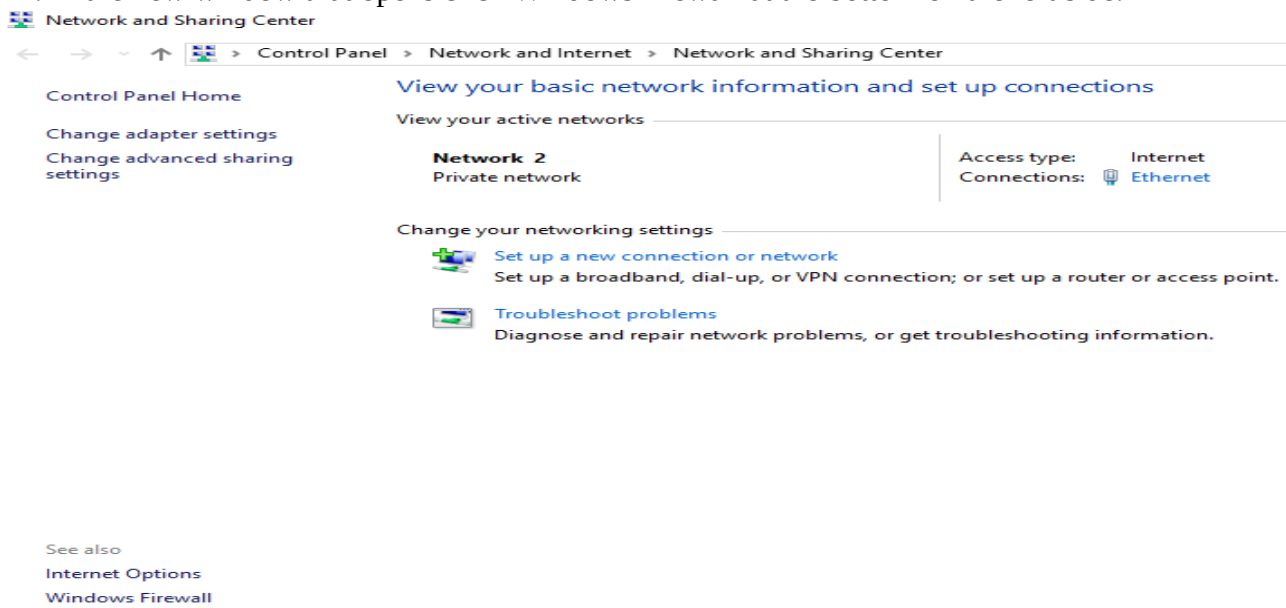
12. Now in the Internet Explorer type **http://localhost:8080/**. The following tomcat page should be displayed.



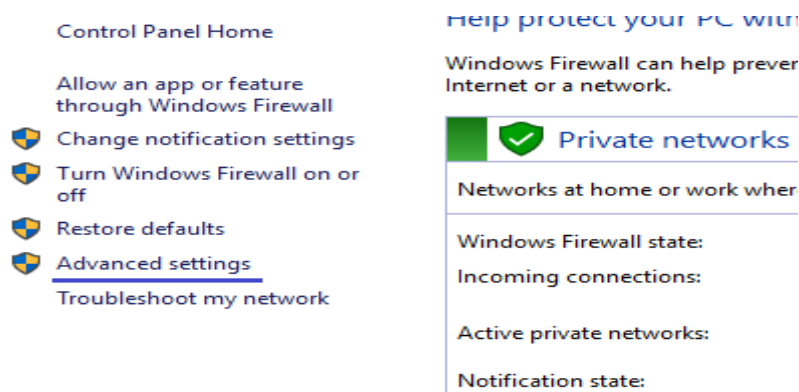
13. Now to open port in Windows firewall right click on the network icon. Click Open Network and Sharing Center.



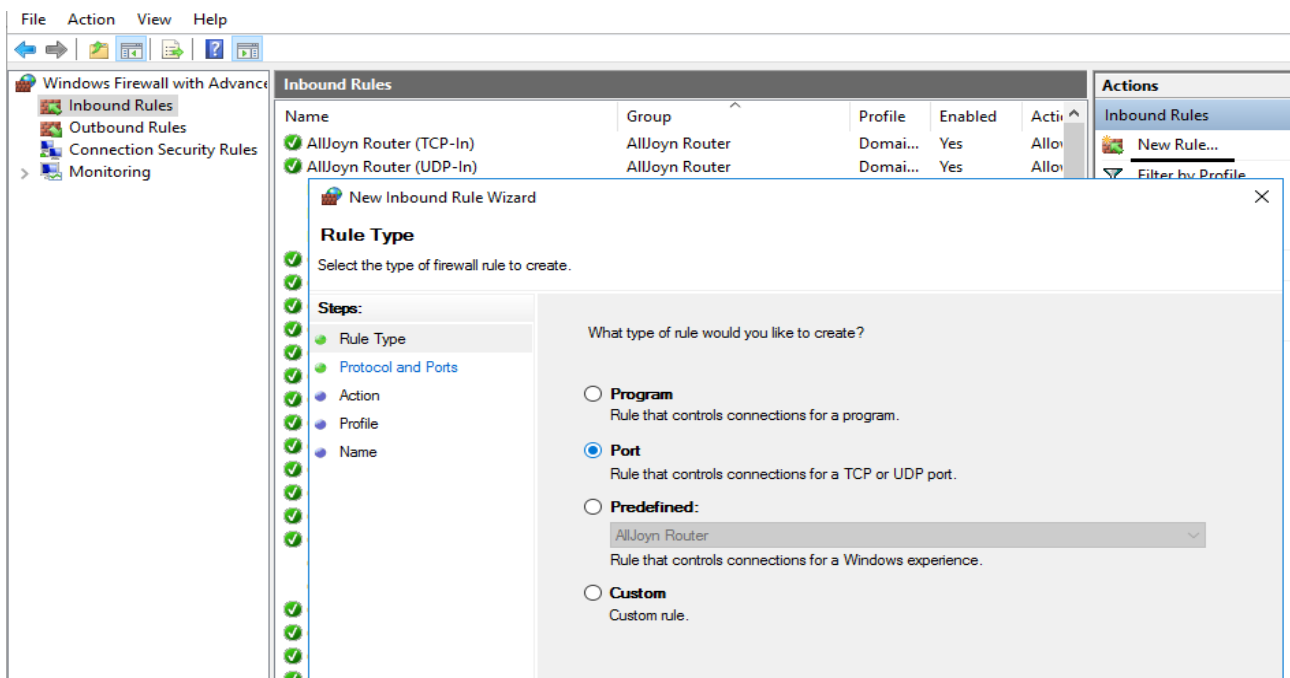
14. In the new window that opens click Windows firewall at the bottom on the left side.



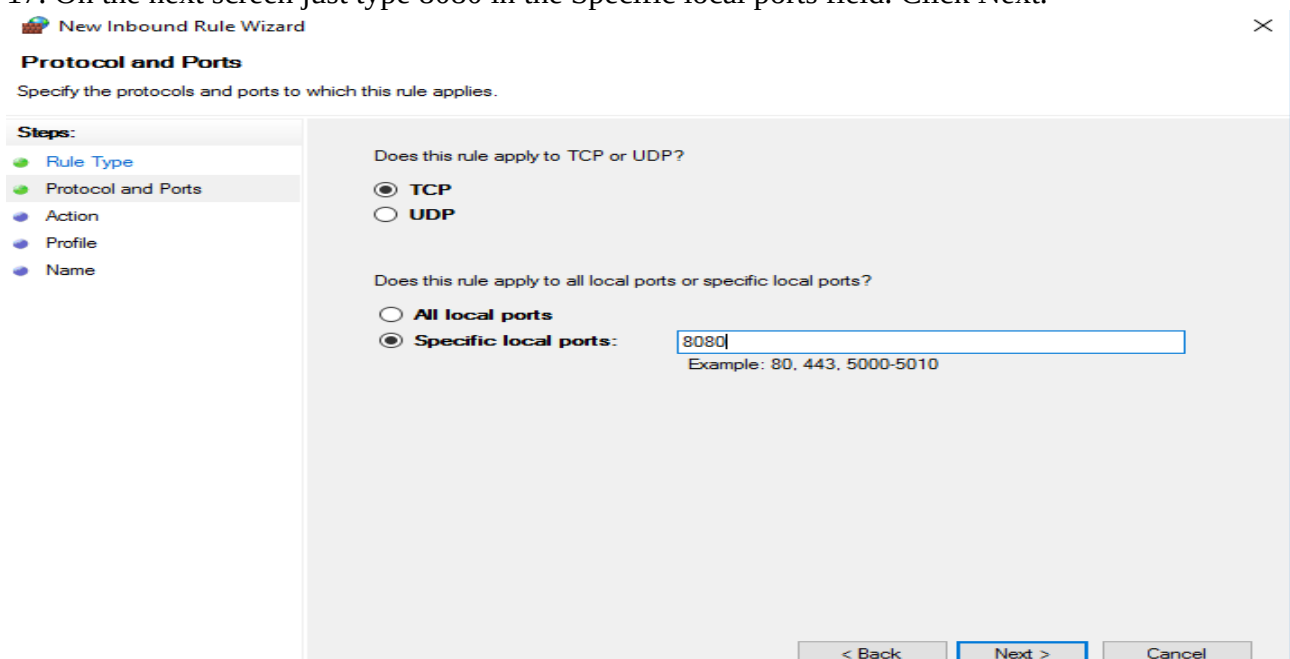
15. In the new Window that opens click Advanced setting on the left side.



16. In the new window that opens click Inbound rules. Then click New rule option on the right hand side. In the new window that opens, click port option and click next.



17. On the next screen just type 8080 in the Specific local ports field. Click Next.



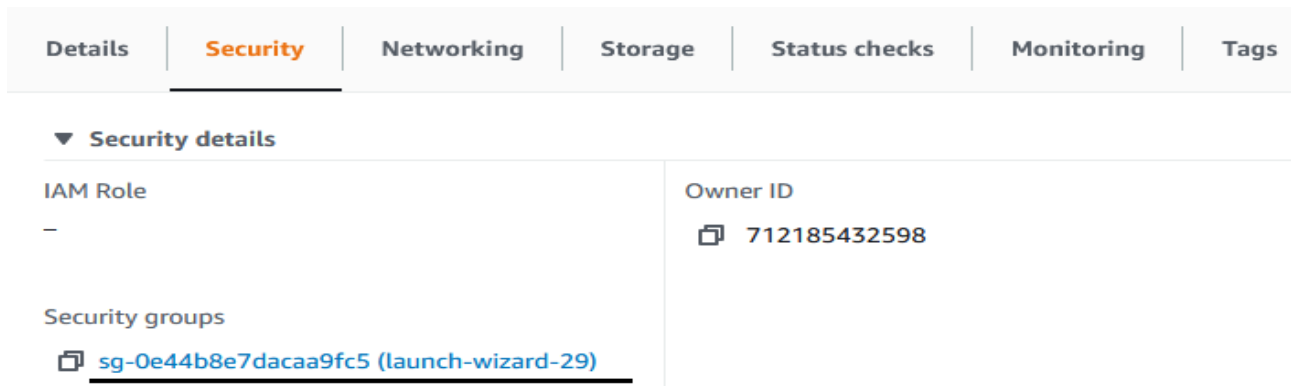
18. On the next screen just click Next.

19. On the next screen also click Next.

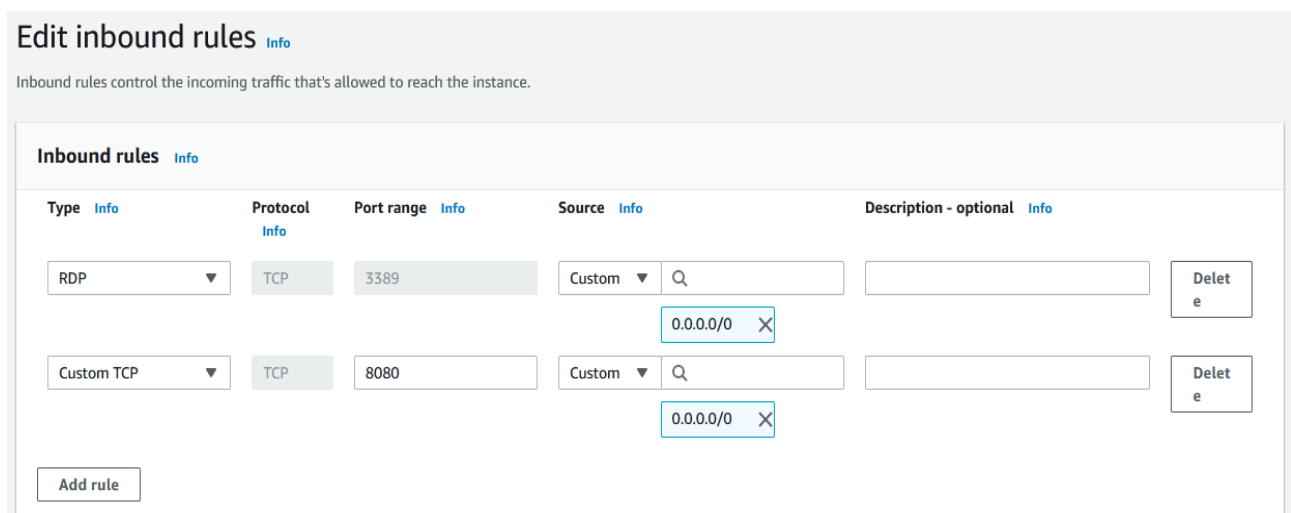
20. On the last screen in the name field type tomcat and click Finish. Then close all windows.

21. Now open this tomcat web server to internet by opening TCP port 8080 in the security group configuration settings associated with this Windows EC2 instance.

22. Minimize the remote desktop screen. Go to the AWS console window. Select the check box in front of the Windows EC2 instance. Below in the security tab click on the security group name.



23. In the new page that opens click the **Edit inbound rules**. Click Add rule. Keep Custom TCP option as it is. In the port range type 8080. In the source keep custom and select 0.0.0.0/0. Then click Save rules button below.



24. Now in the laptop browser type <http://public-ip-windows-instance:8080>. You should be able to see the earlier tomcat page now.

You can close the remote desktop screen anytime. This will just close the connection with the Windows EC2 instance. Anytime you want to connect to this instance again just double click the rdp file downloaded and type the saved password.

After the completion of this practical make sure you terminate the instance.